Below is a **technical specification** for a custom website whose purpose is to compare **baseline vs your solution** on paired videos for **two effects**: **flooding** and **explosion**.

I am going to be blunt about one thing first:

A site where users only “pick the video they like more” is **not enough** if you want results you can trust. You need to control for:

- left/right bias,
- order effects,
- fatigue,
- careless clicking,
- unequal video playback behavior,
- repeated exposure to the same pair,
- and missing “tie / no preference” cases.

So the specification below is designed to produce **honest, analyzable results**, not just pretty survey pages.

---

1. Project purpose

1.1 Objective

Build a web application that presents **paired muted videos** generated from the **same input image**:

- one video from the **baseline solution**
- one video from **your solution**

The application must collect structured user judgments to determine whether your solution is preferred over the baseline.

1.2 Primary research question

At the **same generation budget**, do users prefer videos generated by **your solution** over the baseline?

1.3 Secondary research questions

For which categories does your solution perform better or worse?

- flooding vs explosion
- specific group/category

- difficult scenes vs easy scenes
- identity/structure preservation vs effect realism
- artifact rate / flicker / temporal stability

1.4 Non-goals

The website is **not**:

- a public portfolio page
- a video gallery for browsing
- a content management system for arbitrary media
- a recommender engine
- an annotation tool for frame-by-frame dense labeling

---

2. High-level product design

The website should have **two main zones**:

2.1 Respondent-facing survey application

This is the actual experiment UI for users.

It must:

- show two videos per task
- randomize which side contains baseline vs your method
- collect the user's choice
- optionally collect reason/confidence
- log timestamps and interaction events
- support breaks and progress
- prevent obvious bias and accidental leakage of which model is which

2.2 Admin / researcher dashboard

This is where you manage the study.

It must allow:

- uploading or registering paired videos
- defining study versions
- assigning videos to effect/group
- tracking responses
- exporting results

- monitoring quality-control failures
- freezing a benchmark set before launch

---

3. Core experiment design assumptions

3.1 Dataset structure

You currently have:

- **2 effects**: flooding, explosion
- **19 groups per effect**
- **~5–10 video pairs per group**

Estimated total:

- 2 × 19 × (5–10) = **190 to 380 pairs**

This is too many for one respondent. So the system must support:

- **partial assignment**
- **balanced sampling**
- **multiple study versions**
- **per-respondent subsets**

3.2 Unit of evaluation

The core evaluation unit is a **pairwise comparison task**:

One source image → two generated videos:

- baseline
- your method

The respondent compares those two outputs.

3.3 Experimental rules

The website must enforce:

- same pair shown together only
- baseline/solution identity hidden from the respondent
- left/right placement randomized
- task order randomized or block-randomized
- repeated exposure minimized

---

4. Functional requirements

4.1 Respondent flow

4.1.1 Landing page

Purpose:

- introduce the study briefly
- explain time required
- obtain consent if needed
- optionally collect minimal demographic / experience data

Required elements:

- study title
- one-paragraph explanation
- expected duration
- “Start study” button
- optional consent checkbox
- optional participant code field

Recommended copy theme:

> You will compare pairs of short muted videos generated from the same image. Choose the one that is better overall, or select “No clear preference” if you cannot tell.

Do **not** mention:

- “new method”
- “improved model”
- “baseline”
- “our system”
- “distilled model”

That would bias respondents.

4.1.2 Instructions page

This page must be short and precise.

It should explain what users should evaluate:

- realism of the effect
- temporal stability
- fewer artifacts
- better preservation of the scene/object/person
- overall quality

It should explicitly say:

- do not choose based only on “more dramatic effect”
- videos are muted
- you may replay them
- if there is no meaningful difference, select “No clear preference”

Optional:

- a short practice example with dummy content not used in the real benchmark

---

4.1.3 Eligibility / screening (optional but recommended)

You may want to collect:

- whether the respondent works with generative AI / video / graphics
- whether they are a general viewer or expert

Keep this minimal.

Recommended fields:

- role: general viewer / ML practitioner / artist / video editor / researcher / other
- experience with AI-generated video: none / limited / moderate / strong

This allows subgroup analysis later.

4.1.4 Practice round

Strongly recommended.

Show:

- 2–3 non-study pairs
- teach the interface

- let users understand replay and choice flow

These pairs must not appear in the real evaluation.

4.1.5 Main survey page

This is the most important page in the system.

Each task page must contain:

- Pair ID hidden internally
- Two videos shown in equal-size containers
- Labels only as **A** and **B**
- A response area
- Progress indicator
- Optional replay controls
- Optional keyboard shortcuts
- “Next” button disabled until a response is made

4.1.6 Break / fatigue management page

If a respondent gets more than ~12–15 comparisons, show a short break screen.

Purpose:

- reduce fatigue
- improve answer quality

4.1.7 Completion page

Must display:

- thank-you message
- completion code if needed
- optional summary that does **not** reveal which method was which

---

4.2 Response collection

4.2.1 Primary answer

Each comparison must collect one of:

- Video A is better
- Video B is better
- No clear preference / Tie

This third option is mandatory if you want honest results.

Do **not** force a binary choice.

4.2.2 Secondary answer: reason

Highly recommended.

Allow selecting up to 2 reasons:

- effect looks more realistic
- fewer visual artifacts
- more stable over time
- better preserves the person/object/scene
- better background handling
- stronger but still believable effect
- other

This is very useful for diagnosing where your solution wins or loses.

4.2.3 Confidence rating

Recommended:

- 1 to 5 scale
- “How confident are you in your choice?”

This helps separate strong wins from weak/noisy wins.

4.2.4 Optional “both are poor”

You may optionally add:

- Both are poor, but A is less bad

- Both are poor, but B is less bad

But usually this is too complicated for normal respondents.
Better alternative:

- ask only A / B / tie
- and capture “both are poor” through the reason list or an optional quality rating

---

5. UI/UX specification

5.1 Visual design principles

The site should look:

- clean
- neutral
- not “marketing-like”
- not flashy
- not visually suggestive of which model is better

Use a restrained interface:

- white or dark neutral background
- limited accent color
- no celebratory gradients
- no model branding

This is an evaluation tool, not a demo.

5.2 Layout of comparison page

Recommended desktop layout

Two equal columns:

- left video panel = A
- right video panel = B

Below both panels:

- response buttons
- reason options
- confidence selector
- next button

Top bar:

- study progress
- optional task number
- help button
- replay/reset button

Video panels

Each panel should contain:

- large video display area
- label “A” or “B”
- play/replay button
- optional frame position bar
- loading indicator
- equal dimensions

Avoid

- one panel being more visually prominent
- uneven whitespace
- labels like “option 1” and “option 2” if styling differs
- showing video filenames

Recommended mobile layout

I do **not** recommend mobile as the primary mode.

Explosion/flooding judgments are visually sensitive. Mobile makes:

- flicker harder to detect
- simultaneous comparison harder
- compression/artifacts more confusing

Recommendation:

- desktop-first
- tablet acceptable
- mobile either blocked or allowed with warning

If mobile is allowed:

- stacked layout with a toggle is possible, but worse scientifically
- better to force landscape mode

5.3 Theme

Two reasonable themes:

Light theme

- background: #F7F7F7 or white
- cards: white
- border: subtle gray
- text: dark gray / black
- accent: muted blue

Dark theme

- background: #111 or #181818
- cards: #1E1E1E
- border: #333
- text: near-white
- accent: muted cyan or blue

For video comparison, a **dark theme** is usually better because it reduces glare and lets users focus on the video content.

My recommendation: **default dark neutral theme**.

5.4 Interaction behavior

The interface should support:

- replaying both videos
- synchronized replay
- optional independent replay
- keyboard shortcuts:
    - `1` = choose A
    - `2` = choose B
    - `0` = no preference
    - `R` = replay

    - N = next
- hover or click to enlarge slightly? No. Avoid because it introduces asymmetry.
- a short minimum viewing threshold before answer? Optional, but useful

---

5.5 Progress indicator

Respondents should know where they are.

Show:

- “Comparison 7 of 20”
- progress bar

Do not show:

- group names
- effect names if they can bias expectations

If needed internally, you can store them but not display them.

---

6. Video presentation requirements

6.1 Playback behavior

Videos must be:

- muted
- same duration
- same resolution/aspect ratio
- same fps
- same encoding settings
- same loop behavior

Recommended default:

- autoplay on page load
- muted
- loop once or loop continuously until response
- replay button available

6.2 Synchronization

This matters more than it looks.

Preferred behavior:

- both videos start together automatically
- replay button restarts both from frame 0 together

Why:

- if one video lags or starts later, users compare timing rather than quality

Recommended:

- preload both
- only start when both are ready
- keep them synchronized within a reasonable tolerance

Perfect frame sync is not strictly necessary, but approximate synchronization is very desirable.

6.3 Loading behavior

The page must:

- preload current pair fully if possible
- preload next pair in the background
- show a loading spinner until both videos are ready
- never start one video significantly earlier than the other

6.4 Compression and hosting

This is critical.

You must not let poor encoding distort the study.

Recommended:

- store master files separately
- serve web-optimized files for survey
- same codec/settings for both methods

Suggested delivery formats:

- MP4 (H.264) for broad browser support
- optionally WebM for optimization, but MP4 is safer for universal compatibility

Use the same:

- bitrate target
- CRF / quality settings
- frame rate
- color space assumptions

Do not let one method accidentally look worse because of more aggressive compression.

---

7. Anti-bias and validity features

7.1 Randomization

The system must randomize:

- whether baseline is A or B
- order of tasks per respondent
- order of reason labels if desired

It must also log:

- the mapping for every shown pair

7.2 Tie option

Already mentioned, but it is essential.

Without tie/no preference:

- you will artificially inflate wins
- noisy respondents will create fake signal

7.3 Attention checks

The system should support special hidden QC pairs.

Types:

1. identical-vs-identical pair
    - expected answer: tie/no preference
2. obvious-quality difference pair
    - expected answer: clearly one side

Use a small number:

- 1–2 per respondent if survey length allows

Respondents who fail both repeatedly can be flagged.

7.4 Response-time monitoring

Record:

- time from page render to answer
- time from first playback to answer
- number of replays

Very fast responses are suspicious.
Very slow responses may indicate technical issues.

7.5 Prevent method leakage

The site must not leak:

- filenames
- URLs containing “baseline” or “solution”
- alt text with method names
- visible metadata
- API payloads in easily inspectable frontend state if avoidable

At minimum:

- use neutral UUID-based asset URLs
- map identities server-side

---

8. Study design / sampling logic

8.1 Effects and groups

The site must support:

- 2 effects: flooding, explosion
- 19 groups per effect
- configurable number of pairs per group

Each pair must be tagged with:

- effect
- group
- pair ID
- source image ID
- baseline asset ID
- method asset ID

8.2 Assignment strategy

Do not show every respondent every pair.

Instead use:

- balanced incomplete assignment

Recommended:

- each respondent sees **15–25 pairs**
- each pair gets judged by multiple respondents
- each group receives approximately balanced coverage

8.3 Block design

Recommended study blocks:

- mixed-effect block: flooding + explosion interleaved
  or
- separate effect studies: flooding study and explosion study

I recommend:

- **separate studies per effect** if your respondent count is sufficient
- otherwise one unified platform with effect hidden and balanced internally

Why:

- flooding and explosion have different visual criteria
- mixing them may increase cognitive switching cost

8.4 Stratification

The assignment system should ensure:

- each respondent gets a mixture of easy and hard pairs
- no respondent sees too many pairs from one single group
- per-group sample counts stay balanced globally

---

9. Data model

You need a proper backend schema, not just static files.

9.1 Core entities

Study

Fields:

- `id`
- `name`
- `effect_type` (flooding / explosion / mixed)
- `status` (draft / active / paused / archived)
- `max_tasks_per_participant`
- `instructions_text`
- `created_at`

Group

Fields:

- `id`
- `study_id`
- `name`
- `description`
- `priority`

- `target_votes_per_pair`

Source item

Fields:

- `id`
- `effect_type`
- `group_id`
- `source_image_id`
- `pair_code`
- `difficulty`
- `notes`

Video asset

Fields:

- `id`
- `source_item_id`
- `method_type` (baseline / candidate)
- `storage_url`
- `duration_ms`
- `width`
- `height`
- `fps`
- `codec`
- `checksum`

Pair presentation

This is the randomized version presented to a respondent.
Fields:

- `id`
- `participant_id`
- `source_item_id`
- `left_asset_id`
- `right_asset_id`
- `left_method_type`
- `right_method_type`
- `task_order`
- `is_attention_check`

Participant

Fields:

- `id`
- `session_token`
- `study_id`
- `device_type`
- `browser`
- `started_at`
- `completed_at`
- `quality_flag`

Response

Fields:

- `id`
- `participant_id`
- `pair_presentation_id`
- `choice` (left / right / tie)
- `reason_codes`
- `confidence`
- `response_time_ms`
- `replay_count`
- `created_at`

Interaction log

Fields:

- `id`
- `participant_id`
- `pair_presentation_id`
- `event_type`
- `event_ts`
- `payload_json`

Event types:

- page_loaded
- video_ready_left
- video_ready_right
- autoplay_started
- replay_clicked

- choice_selected
- next_clicked

---

10. Backend/API requirements

10.1 Core API endpoints

Public / respondent endpoints

- `POST /api/session/start`
- `GET /api/session/:id/next-task`
- `POST /api/task/:id/response`
- `POST /api/task/:id/event`
- `POST /api/session/:id/complete`

Admin endpoints

- `POST /api/admin/login`
- `POST /api/admin/studies`
- `POST /api/admin/groups`
- `POST /api/admin/pairs/import`
- `POST /api/admin/assets/upload`
- `GET /api/admin/results/summary`
- `GET /api/admin/results/export`
- `GET /api/admin/qc`

---

11. Admin panel requirements

The admin panel should allow:

11.1 Study creation

- create new study
- choose effect scope
- set number of tasks per respondent

- configure tie option on/off
- configure reason labels
- configure confidence rating on/off

11.2 Pair management

- upload pair metadata CSV
- associate videos with pair IDs
- preview a pair
- mark as active/inactive
- mark as QC pair
- assign group/effect tags

11.3 Recruitment control

- optionally generate invite links
- track how many completions per study
- pause or stop study when enough votes are collected

11.4 Analytics dashboard

At minimum show:

- total respondents
- total completed comparisons
- completion rate
- average time per task
- tie rate
- win rate by effect
- win rate by group
- QC failure rate
- per-pair vote counts

---

12. Frontend technical recommendations

12.1 Recommended stack

For a serious but manageable build:

Frontend

- **Next.js** or **React**

- TypeScript
- Tailwind CSS or a component system
- video handling via native HTML5 `<video>`

Backend

Either:

- Next.js API routes for a unified app
  or
- separate backend in FastAPI / Node.js (Express/Nest)

Database

- PostgreSQL

Storage

- S3-compatible object storage
- Cloudflare R2 / AWS S3 / Backblaze B2 / MinIO

Auth

- admin auth only
- respondents anonymous or token-based

This stack is practical and robust.

12.2 Why not a purely static site

Because you need:

- randomization
- session assignment
- analytics
- QC
- result export
- server-side masking of method identity

A static HTML page is too weak for this.

---

13. Performance requirements

13.1 Functional performance

The site must:

- load the next task quickly
- start both videos with minimal delay
- avoid one-sided buffering
- work stably on normal broadband

13.2 Non-functional performance targets

Suggested targets:

- first task load < 3 seconds on typical connection
- next task transition < 1.5 seconds if preloaded
- video start skew between left/right < 200 ms if possible
- no more than 1% task load failures

---

14. Security and privacy

14.1 Respondent privacy

Unless you truly need identity, do not collect:

- full name
- email
- IP in raw form for long-term storage

If you store IPs for abuse prevention:

- hash or minimize retention

14.2 Admin security

- authenticated admin panel
- rate limiting
- CSRF protection
- secure secret storage
- signed asset URLs if needed

14.3 Asset protection

Do not expose obvious method names in video URLs.
Use:

- opaque filenames / UUIDs
- server-side mapping

This does not make scraping impossible, but it reduces leakage.

---

15. Accessibility and usability

Required

- keyboard navigation
- visible focus states
- readable font sizes
- high contrast controls
- captions are unnecessary because videos are muted and comparison-based, but instructions must be clear
- buttons must be large and easy to click

Nice to have

- “replay both” keyboard shortcut
- “show instructions again” help panel

---

16. Logging and analytics specification

You must log enough to debug bad data.

For each task:

- participant ID
- pair ID
- group
- effect
- mapping baseline↔A/B
- response choice
- reasons
- confidence
- response time

- replay count
- load errors
- browser/device

This is essential for later analysis.

---

17. Result export format

The platform must export:

- CSV
- JSON

CSV should include one row per response with:

- respondent ID
- task order
- effect
- group
- pair ID
- left method
- right method
- choice
- tie flag
- reason tags
- confidence
- response time
- replay count
- QC flag

---

18. Statistical-analysis support features

The platform should directly support computation of:

- overall win rate
- win rate excluding ties
- tie rate
- win rate per effect
- win rate per group
- respondent-level quality filtering
- pair-level vote sufficiency
- confidence-weighted summaries if desired

Optional but good:

- export ready for Bradley–Terry or logistic mixed-effects analysis

---

19. Recommended UX content

Landing page wording

Neutral, short, non-leading.

Instructions

Must explicitly say:

- focus on overall visual quality
- consider realism, stability, and artifacts
- do not choose only based on stronger effect
- use tie/no preference if needed

Response wording

Keep concise.

Example:

- Which video is better overall?
    - A
    - B
    - No clear preference

Then:

- Why?
- Confidence

---

20. Suggested screen structure

Page 1: Welcome

- title
- duration
- consent
- start

Page 2: Instructions

- 4 criteria
- tie option explanation
- continue

Page 3: Practice

- 2 examples

Page 4+: Task pages

- two synchronized videos
- choice panel
- next button

Midpoint break

- short message

Final page

- thank you
- completion code

---

21. Recommended visual behavior of videos

For each task:

- both videos start paused until both are loaded
- then both autoplay together
- loop
- replay button resets both
- user can pause both simultaneously

- optionally independent pause only in advanced mode, but default should be synchronized playback

Do not autoplay one while the other is buffering.

---

22. Quality-control rules

The site should flag:

- too-fast responses
- failure on attention checks
- excessive straight-lining behavior
- zero replays plus extremely fast completion across all tasks
- unusually high tie rate or random clicking pattern

But be careful:

- some good raters are fast
- some good raters never replay

Use flags, not automatic blind deletion.

---

23. Deployment requirements

Recommended environment

- HTTPS only
- domain/subdomain dedicated to the study
- CDN-backed asset delivery
- object storage for videos
- managed PostgreSQL if possible

Recommended hosting

- Vercel / Netlify for frontend
- Railway / Render / Fly.io / VPS / cloud VM for backend
- S3/R2/B2 for video storage

If you want full control, host everything on your own VPS, but CDN-backed storage is still preferable for video delivery.

---

24. Browser support

Target:

- Chrome
- Edge
- Firefox
- Safari recent versions

Minimum:

- synchronized HTML5 video playback support
- autoplay muted support

Autoplay policies vary, but muted autoplay is generally acceptable in modern browsers.

---

25. Acceptance criteria

The website is acceptable when:

Respondent-side

- can complete a session without technical confusion
- sees no model identity leakage
- can compare videos fairly
- can choose A / B / tie
- can replay both videos easily

Research-side

- randomization works
- response data is complete
- exports are clean
- per-group/effect analysis is possible
- attention checks are supported
- study can be paused/resumed
- pairs can be audited

---

26. Recommended version 1 scope

To avoid overbuilding, the first version should include:

Must-have

- landing page
- instructions
- practice round
- paired video task page
- random A/B assignment
- tie option
- reasons
- confidence
- progress bar
- admin import/export
- balanced task assignment
- CSV export
- QC logging

Nice-to-have later

- adaptive sampling
- respondent accounts
- advanced dashboards
- per-frame video scrubbing
- multi-language support
- online statistical summaries

---

27. Design recommendation for your exact case

If this were my project, I would build it like this:

Study structure

- one platform
- two separate studies inside it:
    - Flooding benchmark
    - Explosion benchmark

Session design

- each respondent sees **20 pairs**
- one effect per session
- balanced across groups
- 2 practice pairs
- 1 attention check
- 1 short break after 10 tasks

Task page

- dark neutral theme
- two synchronized muted looping videos
- A/B/tie choice
- 5 reason tags
- confidence 1–5
- keyboard shortcuts

Admin

- CSV pair import
- group/effect tagging
- win-rate dashboard
- export raw response logs

That would be a strong, honest evaluation platform without becoming absurdly complex.

---

28. One important methodological recommendation

Do not call the site a “poll” internally when designing it.

Treat it as a **controlled paired-comparison evaluation system**.

That mindset changes everything:

- the UI becomes neutral instead of persuasive
- logging becomes rigorous instead of minimal
- analysis becomes trustworthy instead of anecdotal

If you want, I can turn this next into one of two concrete deliverables:

1. a **formal SRS-style specification document** with numbered requirements, or
2. a **page-by-page wireframe and database schema** you can directly hand to a developer.